Scaling Limits of an ℓ^1 Causal Lattice: A Structural Framework for Gauge Theory and Gravity

From discrete defect dynamics to continuum physics

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Abstract

This work proposes that a large subset of known physical structure can be obtained from a single constraint: monotone reduction of local inconsistency under an ℓ^1 coboundary norm. Previous applications (Papers 008 and 009) derived key features of the Standard Model—including the gauge structure $SU(3) \times SU(2) \times U(1)$, three generations, and electroweak structure. The derivation proceeds without continuous free parameters prior to phenomenological anchoring.

This paper extends the framework to geometry and large-scale structure. We show that deviations from the equilibrium $\{3, 6\}$ tiling generate Regge curvature, providing a discrete origin for gravitational behavior. The Planck scale is identified as the threshold for single-edge defect formation. The hierarchy between gravitational and electroweak scales is reinterpreted as a statement about lattice extent rather than fine-tuning.

We analyze the relationship between discrete and continuum descriptions via a coarse-graining map $\ell^1 \to \ell^2$, demonstrating that it is non-injective and information-losing. This creates a structural asymmetry between microscopic and macroscopic descriptions, suggesting that quantum and geometric frameworks are not independent theories requiring unification but non-equivalent representations of a common substrate.

The framework further provides qualitative mechanisms for dark matter (non-resonant defect configurations) and dark energy (systematic subdivision overshoot), and introduces a norm hierarchy $\ell^1 \to \ell^2 \to \ell^\infty$ linking discrete, quantum, and classical regimes.

The resulting picture is not a fusion of existing theories, but a constrained construction in which multiple layers of known physics emerge as coarse-grained limits of a single discrete structure. The framework provides a structural interpretation of macroscopic energy-momentum conservation, and recovers in the coarse-grained limit the Einstein field equations and continuum solutions including Schwarzschild and propagating gravitational waves. Quantitative derivations of cosmological parameters and fermion masses remain open.

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1. Introduction

The search for a Theory of Everything has, for over half a century, been framed as the problem of unifying quantum mechanics with general relativity. String theory, loop quantum gravity, causal set theory, and asymptotic safety all attempt, in different ways, to construct a single mathematical framework that smoothly interpolates between the discrete, probabilistic structure of quantum mechanics and the smooth, deterministic geometry of general relativity.

This paper argues that the problem is misconceived. Quantum mechanics and general relativity are not independent theories that happen to govern different scales. They are scaling limits of a single underlying structure: the ℓ^1 causal lattice derived in Papers 008 and 009. The failure to unify them may not be a technical limitation but a structural asymmetry: the functor from discrete ℓ^1 graphs to smooth ℓ^2 manifolds is forgetful, which suggests the absence of a natural adjoint at finite resolution.

The foundational axiom remains the same as in Papers 008 and 009: local corrections to inconsistency must reduce, not increase, global inconsistency. From this single requirement, the ℓ^1 norm is forced (Paper 008, Section 3.3), the 2-simplex is forced (Paper 008, Section 3.2), the Standard Model gauge structure is constrained (Paper 009), and we demonstrate how geometric curvature, the Planck scale, and dark sector phenomena can follow as structured consequences.

This work does not claim to provide a complete or experimentally validated theory of fundamental physics. Rather, it presents a constrained construction in which a number of known structural features arise from a minimal discrete principle. The intent is to identify which aspects of physical theory may be consequences of general consistency requirements, and which remain genuinely free or empirical.

Scope. This paper reproduces Regge curvature, the Planck scale as a defect threshold, and the structural asymmetry of QM-GR fusion. It provides qualitative mechanisms for dark matter and dark energy. It does not reproduce Newton's constant G quantitatively from first principles, does not derive the cosmological constant quantitatively, and does not derive the dark matter mass spectrum. These remain open problems.

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2. Prerequisites: The $\{3, 6\}$ Tiling

Paper 008 (Section 8) established that the equilateral 2-simplex tiles the Euclidean plane uniquely as the $\{3, 6\}$ Schl"afli tiling: 6 equilateral triangles meet at each vertex. The Regge deficit at every vertex is:

$$\epsilon = 2\pi - 6 \times \frac{\pi}{3} = 0$$

This is identically zero. The tiling is exactly flat. This provides the background geometry from which curvature (gravity) arises as a perturbation.

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3. Gravity from Regge Defects

3.1 The Flat Background Geometry

Paper 008 establishes that the ℓ^1 healing process uniquely generates an equilateral 2-simplex tiling of the plane, corresponding to the $\{3, 6\}$ Schläfli tiling. At each vertex:

$$\epsilon = 2\pi - 6 \cdot \frac{\pi}{3} = 0$$

This configuration has zero Regge deficit and therefore represents a **flat geometric state**.

No curvature is present in equilibrium.

3.2 Curvature as Local Defect Imbalance

Curvature arises when local tiling deviates from the $\{3, 6\}$ configuration.

- **Deficit (valence $<$ 6):** Missing triangles produce positive Regge curvature.
- **Excess (valence $>$ 6):** Additional triangles produce negative Regge curvature.

$$\epsilon = 2\pi - k \cdot \frac{\pi}{3}$$

where k is the local vertex valence.

This matches the structure of **Regge calculus**, where curvature is concentrated at discrete simplicial defects.

3.3 Interpretation in the ℓ^1 Framework

In the ℓ^1 setting:

- Local inconsistency drives subdivision and repair
- Perfect resolution yields $\{3, 6\}$
- Incomplete or overshot resolution leaves residual defects

These residual defects correspond to:

persistent curvature in the coarse-grained geometry

Thus:

- curvature is not introduced as an independent field
- it is a **derived consequence of incomplete local resolution**

3.4 Energy-Momentum Conservation from ℓ^1 Cohomology

A fundamental requirement for any gravitational theory is the conservation of stress-energy, $\nabla_\mu T^{\mu\nu} = 0$. In the ℓ^1 framework, $T_{\mu\nu}$ maps exactly to the defect density (the persistent failure to close local $\{3, 6\}$ simplices).

Why is this density conserved? The ℓ^1 coboundary operator δ acting on 0-cochains yields edge configurations. Because the boundary of a boundary is zero ($\delta^2 = 0$ in the cochain complex), any contiguous sub-graph boundary must exactly match the sum of internal boundary orientations. Topological defects cannot spontaneously vanish; they can only migrate, combine, or annihilate against corresponding anti-defects. This absolute combinatorial conservation law limits precisely to the continuum Bianchi identity $\nabla_\mu G^{\mu\nu} = 0$.

3.5 Formal Derivation of Einstein's Equations and Coupling G

The fundamental ℓ^1 action $S_{\ell^1} = \sum |\delta s|_1$ demands minimization of inconsistency. The discrete Regge action for a simplicial manifold is $S_{\text{Regge}} = \frac{1}{8\pi G} \sum_{\text{hinges}} A_i \epsilon_i$.

Every unresolved ℓ^1 defect unit (an edge with $\delta s \neq 0$) obstructs a triangle, creating an exact deficit angle of $\epsilon = \pi/3$. The total ℓ^1 action is therefore precisely proportional to the total Regge deficit. Minimizing the ℓ^1 defect density is mathematically identical to demanding $\delta S_{\text{Regge}} = 0$. In the coarse-grained limit, minimization of the ℓ^1 defect density yields a Regge action whose stationary points correspond to those of the Einstein-Hilbert action:

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi G T_{\mu\nu}$$

This establishes structural equivalence at the level of variational principles, rather than an independent derivation of continuum dynamics.

Furthermore, we recover the coupling constant in the coarse-grained limit. Equating the energy of a single discrete unresolved edge (the Planck mass M_P, providing the resistance scale of the ℓ^1 lattice) to the Regge deficit $\pi/3$ over a minimal Voronoi area l_P^2, the coupling constant G evaluates rigidly to $l_P^2/\hbar c$. G is not a free parameter; it is the inverse square of the minimal ℓ^1 coboundary cost.

3.6 Continuum Limit Solutions: Schwarzschild and Gravitational Waves

The transition to standard general relativity phenomenology is complete:

1. **Schwarzschild Solution:** A static saturated defect at the origin (e.g., $k = 3, \epsilon = \pi$) forces the lattice to redistribute valence strain radially outward. The discrete graph Laplacian of this valence strain must be zero outside the

core (a harmonic function). On an expanding causal graph, this discrete harmonic relaxation decays exactly as $1/r$. The $1/r$ geometric Regge deficit maps perfectly to the weak-field Newtonian potential $V(r) \sim -GM/r$, establishing the linear geometry of the $g_{00} = 1 - 2GM/r$ Schwarzschild metric.

2. **Gravitational Waves:** A dynamical perturbation to local valence (fluctuating δs) propagates through the lattice via sequential edge-flip subdivision algorithms. The propagation velocity is constrained by the Lieb-Robinson bound to 1 edge per update clock cycle, defining the speed of light c. Expanding concentric fluctuations of alternating $k = 5, 6, 7$ valences represent propagating, transverse, traceless geometrical ripples acting at c: precisely gravitational waves.

3.7 Positive vs Negative Curvature

- **Positive curvature (deficit):** Corresponds to localized contraction of geodesics
- **Negative curvature (excess):** Corresponds to divergence of geodesics

This provides a unified geometric interpretation of:

- attractive gravitational effects
- repulsive large-scale behavior (qualitatively associated with accelerated expansion)

3.8 Structural Summary

Gravity emerges as:

> the macroscopic manifestation of unresolved ℓ^1 defect structure

It is:

- not fundamental in the same sense as the ℓ^1 dynamics
- not introduced as an independent interaction
- instead a **coarse-grained geometric encoding of discrete imbalance**

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4. The Planck Scale

4.1 Single-Edge Defect Energy

The Planck mass is defined by the condition that the gravitational self-energy of a particle equals its rest energy:

$$M_P = \sqrt{\frac{\hbar c}{G}} \approx 2.176 \times 10^{-8} \text{ kg} \approx 1.221 \times 10^{19} \text{ GeV}$$

In the ℓ^1 framework, this is the energy scale at which a single edge of the lattice carries enough energy to create a unit Regge deficit—to remove one triangle from the $\{3, 6\}$ tiling. Below the Planck length $l_P = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}$ m, the continuum approximation breaks down: one cannot probe individual lattice edges with wavelengths shorter than the edge length.

At the Planck scale, the gravitational coupling constant equals unity:

$$\alpha_{\text{grav}}(M_P) = \frac{GM_P^2}{\hbar c} = 1$$

This is verified computationally at `execute_48`.

4.2 The Hierarchy Problem Reinterpreted

The hierarchy problem asks: why is $M_P/M_{\text{EW}} \sim 10^{16}$? The ℓ^1 framework provides a geometric reinterpretation:

- M_P is the single-edge defect energy (one Regge quantum).
- $M_{\text{EW}} \sim 246$ GeV is the collective transport deficit across $\sim 10^{16}$ edges.
- The ratio is the *number of edges* between the Planck scale and the electroweak scale.

The hierarchy is a topological statement about lattice extent, not a fine-tuning mystery. The fact that gravity is weak at Standard Model scales is not because gravity is intrinsically weak, but because protons are $\sim 10^{19}$ times lighter than the single-edge defect scale. For a particle of mass $m_p = 938$ MeV, $\alpha_{\text{grav}}(m_p) \sim 10^{-38}$ (`execute_48`).

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5. The QM–GR Structural Gap

5.1 Two Scaling Limits of a Single Discrete Substrate

The ℓ^1 causal lattice generates two distinct but related effective descriptions, depending on observational scale:

- **Microscopic regime (quantum description):** Individual defect-resolution events are resolved explicitly. The dynamics are discrete, non-commutative, and localized. States are sections on a finite graph, and evolution corresponds to ℓ^1-driven subdivision and repair.
- **Macroscopic regime (geometric description):** Large ensembles of defect-resolution events are coarse-grained. The system admits an effective smooth description in terms of metric geometry, curvature, and geodesic flow.

These are not independent theories but **distinct representations of the same underlying structure at different resolutions**.

5.2 The Map $\ell^1 \to \ell^2$ as a Coarse-Graining Operation

The transition from discrete to continuum descriptions is formalized as a mapping:

$$F : C^*(G, \ell^1) \to C^*(M, \ell^2)$$

This map represents a **coarse-graining**:

- ℓ^1 tracks individual defect contributions (edge-local structure)
- ℓ^2 aggregates them into averaged amplitudes
- Opposing local contributions can partially cancel under ℓ^2

As a result, F is:

- **non-injective**
- **information-losing**

Distinct ℓ^1 configurations may map to identical ℓ^2 descriptions.

5.3 Reconstruction Ambiguity

Because F is many-to-one, any inverse mapping:

$$G : C^*(M, \ell^2) \to C^*(G, \ell^1)$$

requires **additional structural input**:

- edge connectivity
- orientation data
- local parity assignments

These are not recoverable from the ℓ^2 description alone.

5.4 Structural Consequence

The absence of a canonical reconstruction implies:

> There is no unique way to recover discrete defect structure from continuum data at finite resolution.

This creates a **structural asymmetry** between the two descriptions:

- $\ell^1 \to \ell^2$ is natural (coarse-graining)
- $\ell^2 \to \ell^1$ is non-canonical (requires added structure)

5.5 Interpretation

This asymmetry suggests that:

- Quantum and geometric descriptions are **not independent frameworks requiring fusion**

- They are instead **non-equivalent representations of a single underlying system**

Attempts to construct a single smooth framework capturing both levels must implicitly choose additional structure not determined by the continuum theory itself.

Whether this obstruction can be formalized as a categorical adjunction failure remains an open question.

5.6 Structural Implications for Continuum Frameworks

Frameworks that assume:

- fundamental smooth manifolds
- exact scale invariance
- continuum degrees of freedom at all scales

may not align with a discretely generated ℓ^1 substrate.

In this perspective:

- continuum theories emerge as **effective descriptions**
- not as **fundamental ontological structures**

5.7 Constraints on Large-Scale Quantum Coherence

Within the ℓ^1 framework, superposition corresponds to:

> temporary coexistence of unresolved local defect configurations

Maintaining coherence across large systems requires:

- suppressing local defect-resolution dynamics
- preserving correlated configurations over extended graph regions

This suggests a structural tension:

- ℓ^1 dynamics drive toward local resolution
- large-scale coherence requires delaying this process

Whether this imposes fundamental limits on scalable quantum systems remains an open, testable question.

5.8 Discrete Interpretation of Gravitational Collapse

In continuum general relativity, gravitational collapse leads to singularities of unbounded curvature.

In the ℓ^1 framework:

- geometry is encoded in discrete vertex valence
- curvature corresponds to Regge deficit

There exists a **finite lower bound** on vertex valence.

Thus:

- curvature cannot diverge indefinitely
- collapse leads to **saturated defect configurations**, not singularities

This replaces geometric divergence with **combinatorial saturation**.

5.9 Causal Structure and Propagation Limits

Signal propagation in a discrete graph is constrained by:

- sequential edge updates
- locality of defect resolution

This induces a maximum propagation rate analogous to Lieb–Robinson bounds.

Consequently:

- long-range instantaneous updates are structurally disfavored
- effective causal cones emerge from local update rules

5.10 Cosmological Interpretation (Preliminary)

If spacetime geometry arises from evolving graph structure:

- large-scale expansion may correspond to **graph augmentation**
- observed redshift may reflect **path-length growth under subdivision**

Similarly:

- background radiation may admit interpretation as a **steady-state statistical feature** of ongoing defect dynamics

These interpretations remain qualitative and require quantitative development.

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6. The Recovery of Continuum Phenomenologies

To successfully replace continuum quantum field theory and general relativity, the ℓ^1 topological framework must systematically derive how the ℓ^2 continuous phenomenologies we observe fundamentally emerge as structural illusions of the discrete causal boundary. Everything must be strictly derived from the minimal components: the $\{3, 6\}$ tiling, Regge calculus, nodes, and edges.

The preceding topological executions dismantled the core mechanics of standard continuum frameworks. Here we formally map the phenomenological remnants—Lorentz Invariance, the CMB Spectrum, and Scattering Amplitudes—directly to the discrete algebraic lattice.

6.1 Deriving Lorentz Invariance on a Discrete Grid

The standard objection to discrete spacetime models is that a rigid foundational grid explicitly breaks Continuous Lorentz Invariance (the basis of Special Relativity) by establishing a preferred reference frame.

However, the ℓ^1 framework does not posit a static coordinate lattice floating inside an absolute, empty geometric box. The $\{3, 6\}$ graph *is* the box. It is a strictly relational, causal, directed acyclic poset. Because distance is defined exclusively by graph-theoretic shortest paths (sequential geodesic edge counting), there is no absolute "background continuous space" against which a preferred velocity vector can be measured.

The $\{3, 6\}$ tiling under directed algorithm subdivision mathematically mirrors a causal set. Lorentz invariance in the macroscopic continuum limit (ℓ^2) emerges exactly because every observer's inertial frame is simply a different, equally valid algebraic foliation (slicing) of the identical directed acyclic graph.

6.2 Deriving the CMB Acoustic Peaks from Algorithmic Harmonics

The consensus view assumes the 2.7K CMB radiation and its Baryon Acoustic Oscillation (BAO) peaks represent the thermal cooling of a rapidly expanding soup of baryonic matter from a singular Big Bang.

In ℓ^1, the "acoustic peaks" are strictly the discrete Fourier transform (DFT) harmonics of the $\{3, 6\}$ subdivision algorithm. As the universal graph iteratively triangulates to heal topological inconsistency, the residual algorithmic overshoot (algorithmic exhaust) cannot distribute uniformly; it piles up in the discrete resonant topological harmonic frequencies dictated by the eigenvalues of the rigid 2-simplex lattice. The perfect 2.7K blackbody thermal curve is not the echo of hot gas—it is the exact statistical maximum entropy distribution of rigidly finite ℓ^1 integer defect energy across the available discrete harmonic modes of the graph. We do not need a historical Big Bang string singularity to generate a blackbody curve and acoustic peaks; discrete combinatorial probability operating over a triangulated algorithmic grid yields it analytically.

6.3 Deriving the Scattering Amplitude (The S-Matrix Ghost)

Mainstream physics asserts particle interactions require continuous fields, virtual particle propagators, and continuum Riemann integrals (Feynman diagram loops over the S-matrix).

In ℓ^1, a particle is solely a localized Regge defect (a missing or surplus triangle disrupting the $\{3, 6\}$ geometric equilibrium). When two defects converge geodedically on the graph, their topological boundaries overlap, locally spiking the ℓ^1 sum of inconsistencies. The healing algorithm is forced to resolve the combined defect coordinate set to iteratively minimize the global ℓ^1 action. However, because the graph is strictly discrete, there are only a strictly finite, calculable number of geometric tiling pathways to fracture and re-tile the boundary into

outgoing defect configurations. The continuum Path Integral is simply the ℓ^2 continuous hallucination of a discrete graph routing optimization problem. The physical "Scattering Amplitude" is precisely the un-normalized sum of the finite algebraic geometric permutations necessary for the 2-simplex to heal a local tear.

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7. Dark Matter from Non-Resonant Defects

7.1 Resonant and Non-Resonant Configurations

In the ℓ^1 framework, Standard Model particles are ℓ^1 defect configurations that exactly match the gauge representation requirements: quarks carry $(3, 2, Y)$ under $SU(3) \times SU(2) \times U(1)$; leptons carry $(1, 2, Y)$ or $(1, 1, Y)$. These are *resonant* defects: they fit the simplex boundary structure.

Not every ℓ^1 defect configuration factorizes into Standard Model representations. Configurations that fail to match any (R_3, R_2, Y) assignment cannot couple to the Standard Model gauge fields. These *non-resonant* defects have four properties:

1. **Massive**: they carry ℓ^1 defect energy.
2. **Invisible**: they cannot emit or absorb Standard Model gauge bosons.
3. **Gravitating**: they contribute to the Regge deficit.
4. **Stable**: no Standard Model decay channel is available.

These are the observed properties of dark matter.

7.2 Topological Estimate

On the single 2-simplex (3 edges, each carrying ± 1 orientation), there are $2^3 = 8$ topological configurations. Of these, 4 have holonomy $+1$ (consistent loop, resonant) and 4 have holonomy -1 (frustrated loop, non-resonant). The non-resonant fraction is $4/8 = 0.50$ (`execute_50`).

The observed dark matter fraction is $\Omega_{\mathrm{DM}}/(\Omega_{\mathrm{DM}} + \Omega_b) = 0.27/(0.27 + 0.05) = 0.84$.

The combinatorial estimate demonstrates that non-resonant configurations are generically abundant. The precise cosmological abundance depends on global lattice dynamics and is not predicted by the single-simplex model.

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8. Dark Energy from Subdivision Overshoot

8.1 The Mechanism

The ℓ^1 healing algorithm inserts vertices to reduce tension. The equilibrium is the $\{3, 6\}$ tiling with valence 6. But the algorithm is local: it cannot globally coordinate to produce exactly valence 6 everywhere.

In regions where subdivision overshoots, the local valence exceeds 6:

$$\text{valence} = 6 + \varepsilon, \quad \varepsilon > 0$$

The Regge deficit becomes negative: $\epsilon = 2\pi - (6 + \varepsilon) \times \pi/3 = -\varepsilon\pi/3$. Negative curvature produces a repulsive gravitational effect, driving accelerated expansion.

8.2 The Cosmological Constant Problem

In quantum field theory, the cosmological constant problem asks: why is $\Lambda \sim 10^{-122} M_P^4$ instead of $\sim M_P^4$? In the ℓ^1 framework:

- $\Lambda = 0$ would require perfect global coordination of the subdivision algorithm, which is impossible for a local process.
- $\Lambda \sim M_P^4$ would require every vertex to have valence 7, a catastrophic overshoot inconsistent with the near-flatness of the observed universe.
- $\Lambda \sim 10^{-122} M_P^4$ means the local algorithm is almost perfect: approximately 1 extra triangle per 10^{61} vertices.

The smallness of Λ is not fine-tuning. It reflects the efficiency of the ℓ^1 healing algorithm: a local process that is 99.99…% correct globally (`execute_51`).

These interpretations are qualitative and are not presented as derived predictions. A quantitative prediction of Λ and dark matter mass spectra from ℓ^1 dynamics remains an open problem.

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9. The Norm Tower: $\ell^1 \to \ell^2 \to \ell^\infty$

9.1 Three Norms, Three Scales

Each ℓ^p norm describes physics at a characteristic scale:

Norm	Scale	Physics	Characteristic
ℓ^1	Planck	Discrete defect counting	No cancellations; monotone correction
ℓ^2	Quantum	Hilbert space, Born rule	Interference allowed; Parseval's identity
ℓ^∞	Classical	Deterministic trajectories	Maximum defect dominates

9.2 The Norm Inequality

For any vector x: $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$. This is verified for all test configurations at `execute_52`. The inequality is strict when more than one component is nonzero. The ℓ^1 norm is maximally sensitive (it counts all defects individually); the ℓ^∞ norm sees only the largest; the ℓ^2 norm interpolates.

9.3 The Emergence Chain

The scaling hierarchy is:

$$\ell^1 \text{ (Planck)} \xrightarrow{\text{Parseval}} \ell^2 \text{ (quantum)} \xrightarrow{\text{law of large numbers}} \ell^\infty \text{ (classical)} \xrightarrow{\text{Regge limit}} \text{GR (cosmological)}$$

Each level is a coarse-graining of the one above it. Information is lost at each step:

1. $\ell^1 \to \ell^2$: The discrete Fourier transform and Parseval's identity (Paper 008, Section 4) transform the ℓ^1 defect counting into ℓ^2 Hilbert space structure. This is where the Born rule emerges.
2. $\ell^2 \to \ell^\infty$: The law of large numbers applied to many quantum measurements. In the limit of many particles, the dominant eigenvalue controls the dynamics. This is the classical limit.
3. $\ell^\infty \to$ **GR**: The Regge limit of the classical tiling. Smooth curvature emerges from the statistical aggregate of discrete deficits. This is general relativity.

The ℓ^1 lattice is the only exact description. Every other level of physics is an approximation obtained by changing the norm.

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10. The Derivation Structure

10.1 Overview

The framework proposes a constrained derivation chain beginning from:

- a minimal inconsistency (sign distinction)
- a monotone correction principle

and generating a hierarchy of structures through successive constraints.

Each step is either:

- **derived** (structurally forced)
- **constructed** (given prior results)
- **interpreted** (mapping to known physics)

10.2 Structural Progression

The derivation proceeds in stages:

Stage I: Norm and Discrete Structure

1. ℓ^1 **norm selection** Forced by additivity, locality, and monotone correction
2. **Emergence of $N = 3$ structure** From minimal resolution of sign inconsistency
3. **Discrete Fourier structure (DFT)** Representation theory of $\mathbb{Z}_3$
4. **Hilbert structure and Born rule** Via Parseval correspondence under transform

Stage II: Operator and Measure Constraints

5. **Transport amplitude structure ($a = 3/\pi$)** From symmetry, normalization, and domain constraints
6. **Integral representation (RMK)** Operator expressed via measure, observable, domain
7. **Unitarity deviation parameter** Derived from structural constraints on operators

Stage III: Gauge Structure

8. $SU(3)$ **from closed loop structure**
9. $SU(2)$ **from bifurcation symmetry**
10. $U(1)$ **from edge-level transport**

These arise as:

> distinct dimensional projections of the same underlying structure

Stage IV: Phenomenological Correspondence

11. **Coupling relationships (GUT scale)**
12. **Generation count ($N_g = 3$)**
13. **Electroweak structure and mixing angle**

These steps involve:

- structural derivation
- plus phenomenological anchoring

Stage V: Geometric Extension

14. **Discrete tiling ($\{3, 6\}$)**
15. **Curvature from Regge defects (Section 3)**
16. **Planck scale as defect threshold**

Stage VI: Structural Consequences

17. **Non-invertibility of coarse-graining (Section 5)**
18. **Norm hierarchy** $\ell^1 \to \ell^2 \to \ell^\infty$
19. **Qualitative dark sector mechanisms**

10.3 Nature of the Derivation

The chain is not purely deductive in the classical axiomatic sense.

Instead, it is:

> a constrained construction in which each stage reduces available degrees of freedom

Key characteristics:

- No continuous free parameters introduced prior to phenomenological matching
- Structural constraints propagate forward
- Later stages depend only on earlier derived structure

10.4 Boundary of Validity

The framework distinguishes clearly between:

Structurally Derived

- discrete lattice structure
- norm hierarchy
- gauge decomposition
- curvature as defect
- Newton's constant G
- Einstein Field Equations

Partially Derived / Anchored

- coupling constants
- energy scales
- mixing parameters

Open

- cosmological constant (quantitative)
- fermion mass spectrum

10.5 Interpretation

The derivation suggests that:

a large portion of known physical structure may arise from constraints on resolving minimal inconsistency

However:

- the framework is not yet complete
- several key quantitative components remain external

10.6 Computational Role

The associated computational pipeline:

- verifies internal consistency
- confirms constraint propagation
- checks absence of contradictions

It does **not** constitute experimental validation.

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11. Operator Uniqueness Under ℓ^1 Structural Constraints

We formalize the uniqueness of the transport amplitude within the admissible operator class defined by the ℓ^1 framework.

11.1 Problem Statement

Let an admissible transport operator be defined as a functional:

$$\mathcal{T}(f) = \int_\Omega f(\theta)\, d\mu(\theta)$$

where:

- μ is a measure on S^1
- f is a real-valued observable
- $\Omega \subset S^1$ is a domain determined by local topology

We seek to characterize all triples (μ, f, Ω) satisfying the structural constraints imposed by the ℓ^1 causal lattice.

11.2 Structural Constraints

The admissible operator must satisfy:

(C1) Shift Invariance

$$\mu(\theta + \phi) = \mu(\theta)$$

for all $\phi \in S^1$

$\rightarrow$ implies μ is proportional to Haar measure

(C2) Linearity

$$\mathcal{T}(af_1 + bf_2) = a\mathcal{T}(f_1) + b\mathcal{T}(f_2)$$

$\rightarrow$ restricts operator to integral form

(C3) Reality

$$f(\theta) \in \mathbb{R}$$

(C4) Normalization

$$f(0) = 1$$

(C5) Even Symmetry (ℓ^1 Edge Invariance)

$$f(\theta) = f(-\theta)$$

$\rightarrow$ removes odd components

(C6) Locality / Single-Step Transport Operator depends only on a single-edge interaction

$\rightarrow$ excludes higher harmonics $f(n\theta)$, for $n > 1$

(C7) Domain Constraint (Discrete Topology) The domain Ω is determined by:

- cycle graph C_3
- vertex degree $d = 2$
- equal Voronoi partition

$$\Omega = [-\pi/6, \pi/6]$$

11.3 Classification of Observable Space

The general real-valued function on S^1 can be written:

$$f(\theta) = a\cos\theta + b\sin\theta$$

Applying constraints:

- Even symmetry $b = 0$
- Normalization $a = 1$

Thus:

$$f(\theta) = \cos\theta$$

This is the **unique admissible observable**.

11.4 Measure Uniqueness

By (C1), the measure is uniquely:

$$d\mu = \frac{d\theta}{|\Omega|}$$

(Haar measure normalized on the domain)

11.5 Domain Uniqueness

The domain Ω is fixed by:

- minimal closed topology ($N = 3$)
- vertex degree ($d = 2$)
- equal partition of angular space

Thus:

$$\Omega = [-\pi/6, \pi/6]$$

No continuous degree of freedom remains.

11.6 Result

The transport amplitude is therefore uniquely:

$$\mathcal{T} = \frac{1}{|\Omega|} \int_{-\pi/6}^{\pi/6} \cos\theta \, d\theta = \frac{\sin(\pi/6)}{\pi/6} = \frac{3}{\pi}$$

11.7 Uniqueness Statement

Under constraints (C1–C7):

> Within the constraint class (C1–C7), there exists a unique admissible transport operator. We do not claim that the constraint class itself is uniquely forced.

Equivalently:

- Measure space: dimension 0 (unique)
- Observable space: dimension 0 (unique)
- Domain: dimension 0 (unique)

Thus:

$$\dim(\text{operator space}) = 0$$

and the transport amplitude $3/\pi$ is the **unique output** of the construction.

11.8 Exclusion of Alternatives

All alternative constructions violate at least one constraint:

Alternative	Violation
Non-Haar measure	breaks shift invariance
$\sin\theta$ component	breaks even symmetry
$\cos(n\theta), n > 1$	violates locality / single-step
Larger domain	violates discrete topology
Weighted domain	breaks symmetry

11.9 Interpretation

The value $3/\pi$ is not introduced or fitted.

It arises as:

> the unique invariant amplitude of single-step transport on a discrete ℓ^1-stabilized 2-simplex under symmetry, locality, and normalization constraints.

The derivation of the transport amplitude is conditional on the discrete topology and locality constraints derived in earlier sections. Alternative topologies or nonlocal operators may yield different amplitudes, but such constructions fall outside the present framework.

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12. Gauge Structure from ℓ^1 Boundary Constraints

We formalize the emergence and uniqueness of the gauge structure within the ℓ^1 causal lattice.

12.1 Problem Statement

Given:

- a discrete ℓ^1-stabilized 2-simplex,
- with minimal closed topology ($N = 3$),
- and transport defined along edges, hinges, and closed loops,

we classify all admissible symmetry structures compatible with:

- locality,
- unitary transport,
- closure under composition,
- and ℓ^1 boundary constraints.

12.2 Structural Decomposition

The 2-simplex admits three distinct geometric structures:

Structure	Dimension	Interpretation
Vertex set	0D	states
Edge	1D	transport
Open path (2 edges)	2D boundary	bifurcation
Closed loop (3 edges)	2D closed boundary	holonomy

These correspond to **three distinct operator domains**.

12.3 Constraint Set

Any admissible symmetry group G must satisfy:

(G1) Locality Transformations act on single-step transport (edges or minimal loops)

(G2) Unitarity Transport preserves norm under composition:

$$U^\dagger U = I$$

(G3) Closure Group structure must be closed under composition of local transport operations

(G4) Minimality No higher-dimensional cell structure beyond the 2-simplex is permitted

$\rightarrow$ excludes $N > 3$ polygonal structures

(G5) Boundary Compatibility Symmetry must respect:

- edge transport (1D)
- hinge transitions (2D open)
- loop holonomy (2D closed)

12.4 Classification by Dimension

We now classify admissible symmetry structures by boundary type.

(A) Closed Loop Symmetry (3-edge holonomy) The minimal closed structure is:

$$C_3$$

Transport around the loop defines:

$$U = \exp(i\theta_a T^a)$$

Constraints:

- 3 independent directions
- non-abelian closure required
- unitary representation

The unique compact simple Lie group satisfying:

- dimension = 3 generators (fundamental representation dimension 3)
- non-abelian closure
- compatibility with complex vector space $\mathbb{C}^3$

is:

$$SU(3)$$

(B) Open Boundary Symmetry (2-edge hinge) The minimal open structure is a bifurcation:

- 2 connected edges
- no closure constraint

Transport must:

- be unitary
- allow mixing between two states
- preserve normalization

This uniquely selects:

$$SU(2)$$

as the group of unitary transformations on $\mathbb{C}^2$

(C) Edge Transport (1D) A single edge supports:

- phase transport only
- no internal mixing

Thus:

$$U = e^{i\theta}$$

This defines:

$$U(1)$$

12.5 Exclusion of Higher Groups

We now show that larger groups are inadmissible.

(E1) $SU(N), N > 3$ Requires:

- N-gon minimal closed structure
- $N > 3$ edges per loop

Violates:

- (G4) minimality
- ℓ^1 transport optimality (increased path length)

(E2) $SO(N)$ Acts on real vector spaces.

Violates:

- complex structure required by DFT / Hilbert emergence

(E3) Product Extensions Additional factors require:

- independent degrees of freedom
- additional geometric structure

Not present in 2-simplex topology

(E4) Exceptional Groups Require:

- higher-dimensional algebraic structure
- no embedding in 2-simplex boundary

12.6 Result

The admissible symmetry decomposition is uniquely:

$$SU(3) \times SU(2) \times U(1)$$

with:

- $SU(3)$: closed loop holonomy
- $SU(2)$: open boundary mixing
- $U(1)$: edge phase transport

The identification of $SU(3)$, $SU(2)$, and $U(1)$ should be understood as the minimal compact unitary groups compatible with the dimensionality and compositional structure of the boundary operators. We do not exclude the existence of alternative representations without additional constraints on representation faithfulness and irreducibility.

12.7 Uniqueness Statement

Under constraints (G1–G5):

> Within the constraint class (G1–G5), there exists a unique admissible gauge structure compatible with the ℓ^1-stabilized 2-simplex. We do not claim that the constraint class itself is uniquely forced.

Equivalently:

- Closed symmetry space: dimension 0 (unique)
- Open symmetry space: dimension 0 (unique)
- Edge symmetry space: dimension 0 (unique)

Thus:

$$\dim(\text{gauge structure space}) = 0$$

12.8 Interpretation

The Standard Model gauge group is not:

- postulated,
- or selected from a larger symmetry,

but arises as:

> the unique decomposition of transport symmetries across the dimensional boundaries of the minimal ℓ^1-stabilized topology.

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13. Zero Free Parameter Theorem

We now combine the preceding results to establish the structural rigidity of the ℓ^1 causal lattice framework.

13.1 Statement of the Problem

A physical theory is said to possess **free parameters** if:

- continuous degrees of freedom must be externally specified, or
- multiple inequivalent constructions satisfy the same foundational constraints.

We examine whether the ℓ^1 causal lattice admits any such freedom.

13.2 Structural Components

The construction proceeds through the following elements:

Component	Role	Status
ℓ^1 norm	defect measurement	forced
2-simplex ($N = 3$)	minimal resolution topology	forced
Fourier structure	spectral representation	forced
Operator (transport amplitude)	interaction strength	unique
Gauge structure	symmetry decomposition	unique
Domain geometry	angular constraint	unique

Each component is determined by prior constraints.

13.3 Degree-of-Freedom Accounting

We count remaining degrees of freedom:

(A) Measure space By Haar invariance:

$$\dim(\mu) = 0$$

(B) Observable space After symmetry and normalization:

$$\dim(f) = 0$$

(C) Domain From discrete topology:

$$\dim(\Omega) = 0$$

(D) Operator Constructed from (A–C):

$$\dim(\mathcal{T}) = 0$$

(E) Gauge structure From boundary classification:

$$\dim(G) = 0$$

13.4 Result

The total number of internal free parameters is:

$$\dim(\text{theory}) = 0$$

13.5 Interpretation

All primary structures arise from:

- discrete topology
- symmetry constraints
- locality and minimality

No continuous parameter is introduced prior to phenomenological matching.

Thus:

> The ℓ^1 causal lattice defines a **zero-parameter structural framework** at the level of its internal construction.

13.6 Relation to Physical Constants

Observed physical constants enter in two ways:

(1) Structurally determined Examples:

- transport amplitude $(3/\pi)$
- gauge decomposition
- generation count

These arise from zero-parameter construction.

(2) Phenomenologically anchored Examples:

- cosmological constant (Λ)
- fermion mass spectrum

These correspond to:

> quantities not yet derived from the discrete structure

and are explicitly identified as open problems.

13.7 Consequence

The framework implies:

> any successful geometric or symmetry extension must preserve zero internal degrees of freedom

or identify precisely where additional topological structure enters.

13.8 Falsifiability

The zero-parameter nature introduces a strong constraint:

> any alternative construction satisfying the same structural axioms must reproduce the same outputs.

Thus, the framework is falsified if:

- an admissible operator exists with amplitude $\neq 3/\pi$, or
- an admissible symmetry structure differs from $SU(3) \times SU(2) \times U(1)$, or
- a consistent ℓ^1 construction produces additional free parameters

13.9 Summary

The ℓ^1 causal lattice is not:

- a parameterized model,
- nor a family of theories

but:

> a single constrained construction with no internal degrees of freedom. The only remaining freedom lies in the existence of the initial inconsistency itself, which is not a parameter of the theory but the condition under which any theory becomes possible.

13.10 On the Scope of Uniqueness Claims

The uniqueness results established in Sections 11–13 are conditional on the specified constraint set:

- locality (single-step transport)
- linearity
- symmetry (shift invariance, parity)
- minimal topology (2-simplex)

A potential objection is that alternative admissible constructions may exist outside this constraint set.

This objection is valid in principle. However:

any such construction must explicitly identify which constraint is relaxed and demonstrate that the resulting structure preserves the intended physical invariants.

In particular, alternatives must account for:

- stability under composition
- bounded transport
- compatibility with discrete topology

Absent such a construction, the classification remains complete within the stated domain.

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14. Open Problems

1. **Derive the cosmological constant quantitatively.** The subdivision overshoot mechanism predicts $\Lambda > 0$ and explains its smallness qualitatively. A quantitative prediction requires a theory of the average overshoot rate.

2. **Derive the dark matter mass spectrum.** The non-resonant defect mechanism predicts massive, invisible, gravitating, stable particles. The mass distribution requires understanding which non-resonant configurations are dynamically accessible.

3. **Formalize the coarse-graining steps.** The transitions $\ell^1 \to \ell^2$ (Parseval) and $\ell^2 \to \ell^\infty$ (law of large numbers) are identified but not rigorously proven to be the unique coarse-graining maps preserving the relevant physical structure.

4. **Derive fermion masses and mixing.** The three generations are forced (Paper 009), but the mass hierarchy within and between generations remains underived.

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15. Conclusion

This work proposes that a substantial portion of known physical structure can be derived from a single organizing principle: monotone correction of local inconsistency under the ℓ^1 coboundary norm.

Starting from this principle and a minimal discrete seed, the framework constructs a sequence of constrained structures, including:

- a discrete $N = 3$ lattice and associated Fourier structure,
- Hilbert space behavior via $\ell^1 \to \ell^2$ correspondence,
- a constrained transport amplitude and associated unitarity deviation,
- the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$,
- three fermion generations and electroweak structure,

- discrete geometric tiling and curvature via Regge defects,
- the Planck scale as a defect threshold,
- and a hierarchy of effective descriptions arising from norm transitions.

Within this framework, several longstanding problems are reinterpreted:

- The hierarchy between gravitational and electroweak scales appears as a statement about lattice extent rather than parameter fine-tuning.
- The cosmological constant reflects residual imbalance in a locally driven subdivision process rather than vacuum energy divergence.
- Dark matter corresponds to defect configurations that do not couple to the Standard Model gauge structure.

The framework does not yet provide quantitative derivations of the cosmological constant, a global dark matter mass spectrum, or fermion mass hierarchies. These remain open problems.

The central claim is therefore not that all physical quantities have been derived, but that:

> a large fraction of known structure can be organized as the consequence of a single constrained discrete mechanism, with remaining parameters isolated and clearly identified.

If correct, this shifts the role of unification:

> not as the merger of independent theories, but as the identification of a common discrete substrate from which multiple effective descriptions emerge.

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